



Machine learning based peri-surgical risk calculator for abdominal related emergency general surgery: a multicenter retrospective study

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Background: Currently, there is a lack of ideal risk prediction tools in the field of emergency general surgery (EGS). The American Association for the Surgery of Trauma recommends developing risk assessment tools specifically for EGS-related diseases. In this study, we sought to utilize machine learning (ML) algorithms to explore and develop a web-based calculator for predicting five perioperative risk events of eight common operations in EGS.

Method: This study focused on patients with EGS and utilized electronic medical record systems to obtain data retrospectively from five centers in China. Five ML algorithms, including Random Forest (RF), Support Vector Machine, Naive Bayes, XGBoost, and Logistic Regression, were employed to construct predictive models for postoperative mortality, pneumonia, surgical site infection, thrombosis, and mechanical ventilation > 48 h. The optimal models for each outcome event were determined based on metrics, including the value of the Area Under the Curve, F1 score, and sensitivity. A comparative analysis was conducted between the optimal models and Emergency Surgery Score (ESS), Acute Physiology and Chronic Health Evaluation II (APACHE II) score, and American Society of Anesthesiologists (ASA) classification. A web-based calculator was developed to determine corresponding risk probabilities.

Result: Based on 10 993 patients with EGS, we determined the optimal RF model. The RF model also exhibited strong predictive performance compared with the ESS, APACHE II score, and ASA classification. Using this optimal model, the authors developed an online calculator with a questionnaire-guided interactive interface, catering to both the preoperative and postoperative application scenarios.

Conclusions: The authors successfully developed an ML-based calculator for predicting the risk of postoperative adverse events in patients with EGS. This calculator accurately predicted the occurrence risk of five outcome events, providing quantified risk probabilities for clinical diagnosis and treatment.

Keywords: abdomen, emergency general surgery, machine learning, risk prediction

Introduction

In the past decade, acute care surgery (ACS) has rapidly developed as an emerging specialty^[1], comprising subspecialties of emergency general surgery (EGS), trauma surgery, and surgical

critical care^[2]. The burden of EGS has increased in recent years. Epidemiological data from 2001 to 2010 showed a 28% increase in hospitalizations for EGS in the United States, with over 27 million annual hospitalizations, of which ~28.8% (approximately 7.97 million cases) required emergency surgical

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